BOOTSTRAP

James Doss-Gollin

CEVE 543

2026-09-11

Why uncertainty matters

The bootstrap

Resampling schemes

RETURN-LEVEL UNCERTAINTY GROWS IN THE TAIL

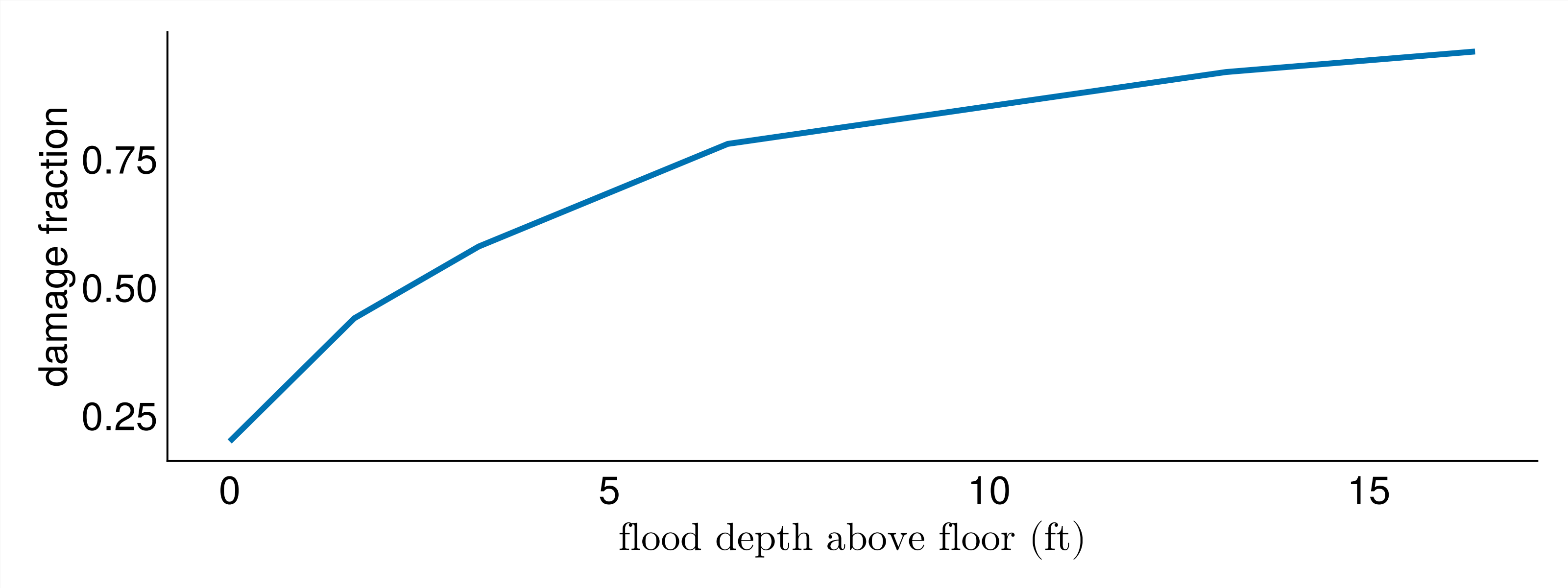
Wednesday's MLE gave one $\hat{\theta}$ and one return-level curve. A different $\hat{\theta}$ gives a different curve, and the curves diverge at long return periods.

Note

Figure shown in class: Return periods for Sewells Point, VA (Doss-Gollin & Keller, 2023)

Depth-Damage Curve

Europa depth-damage function (Doss-Gollin & Keller, 2023):

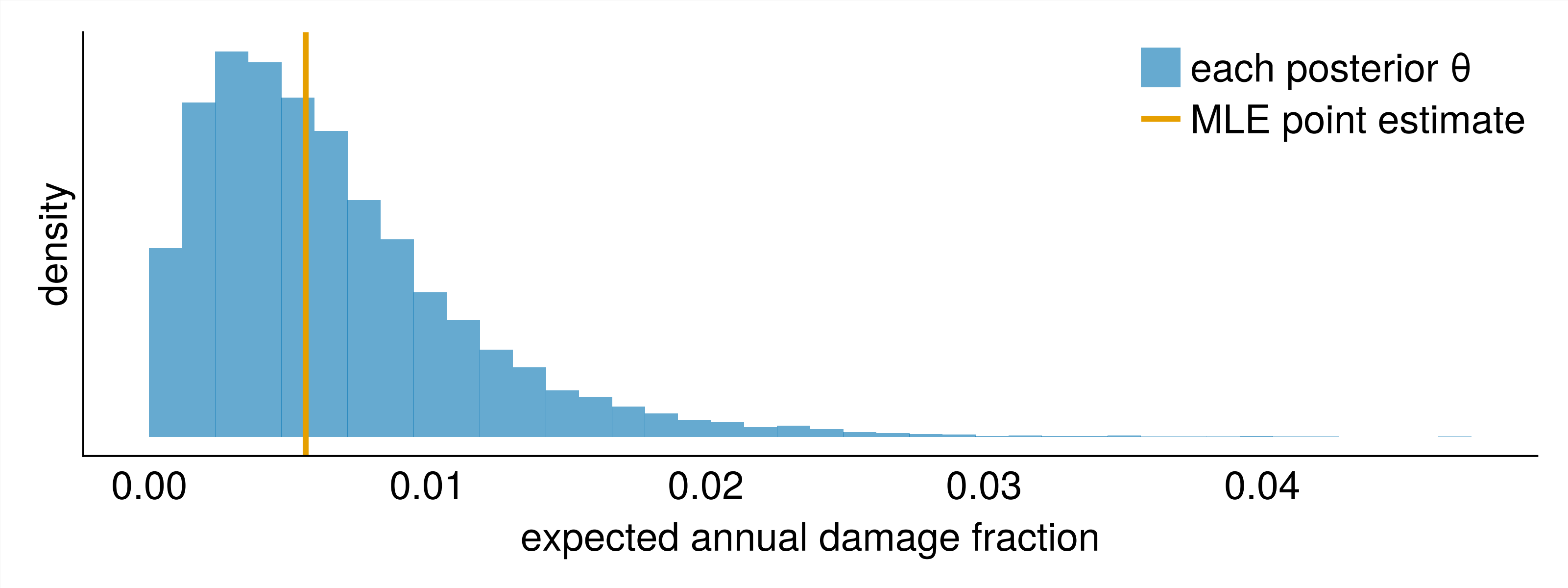


HOW MUCH DOES PARAMETER UNCERTAINTY CHANGE THE DAMAGE?

Expected annual damage for a house at 7 ft elevation, Sewells Point annual maxima.

- **MLE**: one $\hat{\theta}$, one expected annual damage (vertical line)
- **Posterior**: each draw of θ from Wednesday's posterior gives a different expected annual damage (histogram)

HOW MUCH DOES PARAMETER UNCERTAINTY CHANGE THE DAMAGE?



Even modest uncertainty in the return level produces large uncertainty in expected damage, because the damage curve is steep in the tail.

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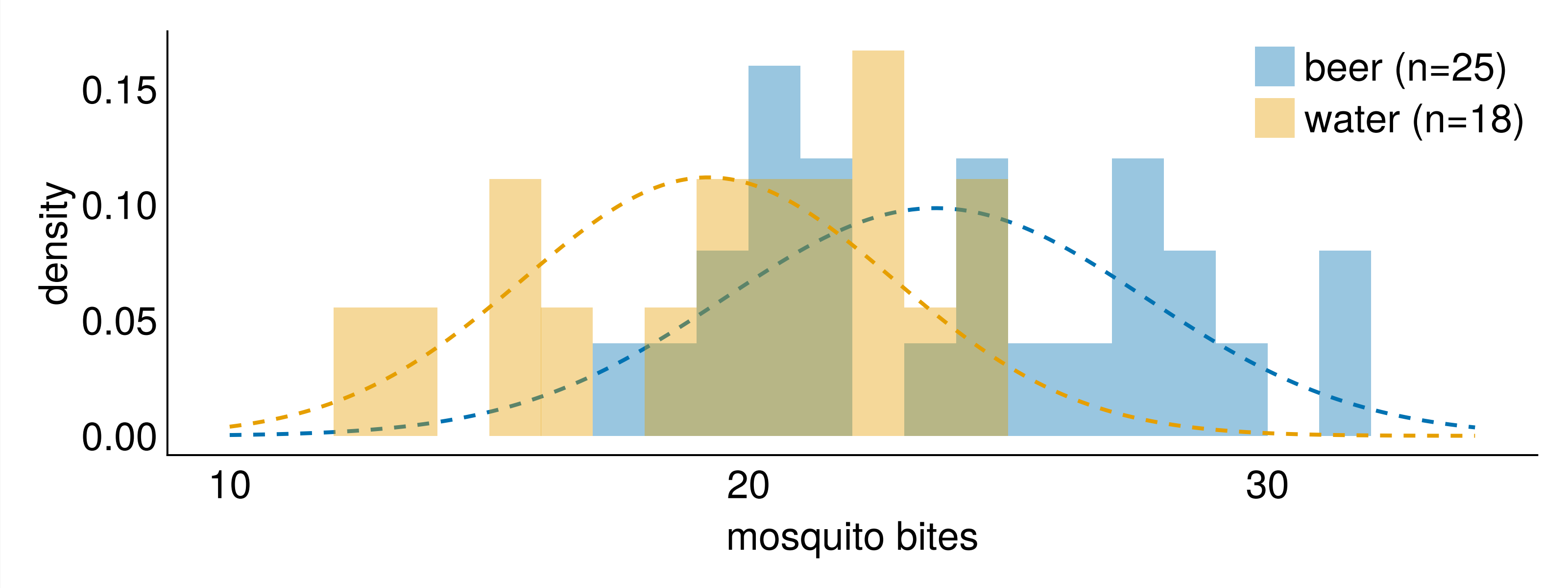
RESAMPLING FROM DATA: THE MOSQUITOES EXPERIMENT

Lefèvre et al. (2010) counted mosquito bites on 25 beer drinkers and 18 water drinkers. **Skeptic:** the difference is chance and the group labels are arbitrary.

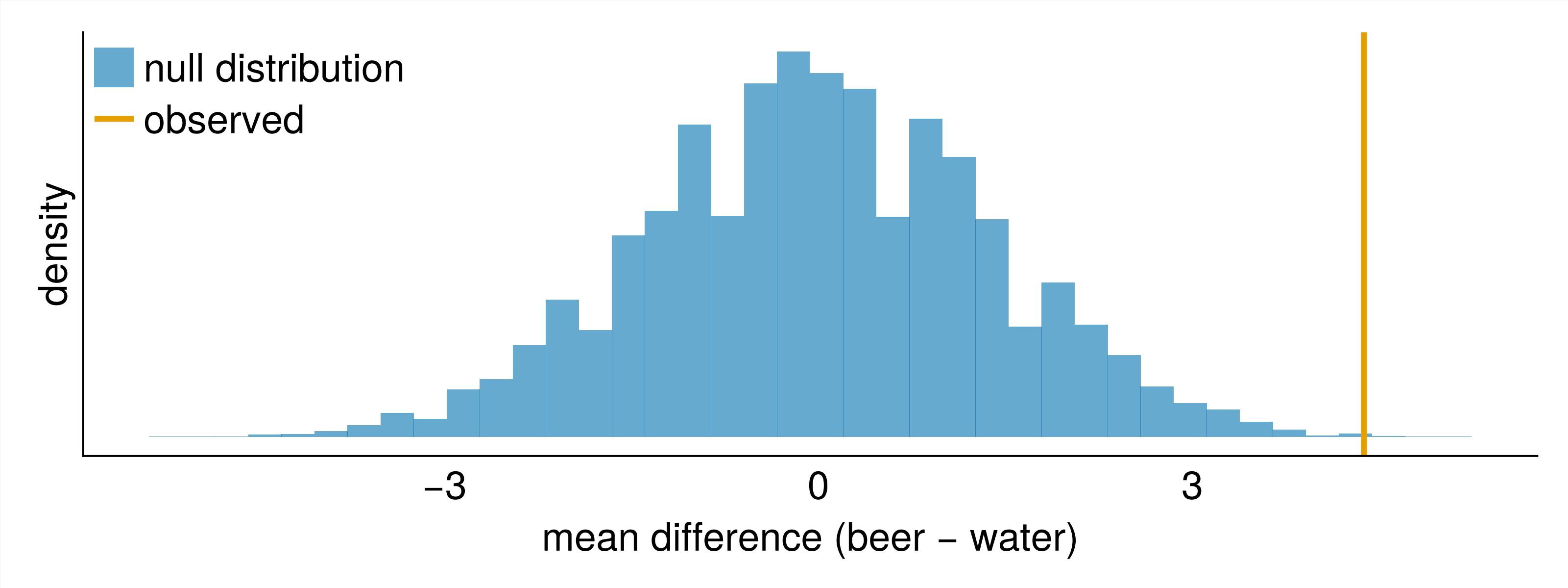


THE PERMUTATION TEST

If the labels are arbitrary, shuffle them and recompute. Each shuffle gives a mean difference under the null hypothesis:

```
function shuffled_difference(y1, y2)
    pooled = shuffle(rng, vcat(y1, y2))
    return mean(pooled[1:length(y1)]) - mean(pooled[(length(y1) + 1):end])
end
null_diffs = [shuffled_difference(beer, water) for _ in 1:10_000]
p_value = mean(null_diffs .>= observed_diff)
```

THE PERMUTATION TEST




```
p-value: 0.0006
```

THE BOOTSTRAP IDEA

The permutation test shuffled labels to test a null hypothesis. The bootstrap resamples the data **with replacement** to estimate uncertainty in $\hat{\theta}$ (Efron & Tibshirani, 1994).

Drawing n values with replacement from your data treats the sample as if it were the population. Some observations appear twice, others drop out, and the estimate changes each time.

THE PROCEDURE

1. Draw a **resample**: n values from the original data, with replacement
2. Compute $\hat{\theta}^*$ on the resample
3. Repeat B times to get $\hat{\theta}^{*1}, \ldots, \hat{\theta}^{*B}$

A 95% confidence interval is the 2.5th and 97.5th percentiles of the $\hat{\theta}^*$ values.

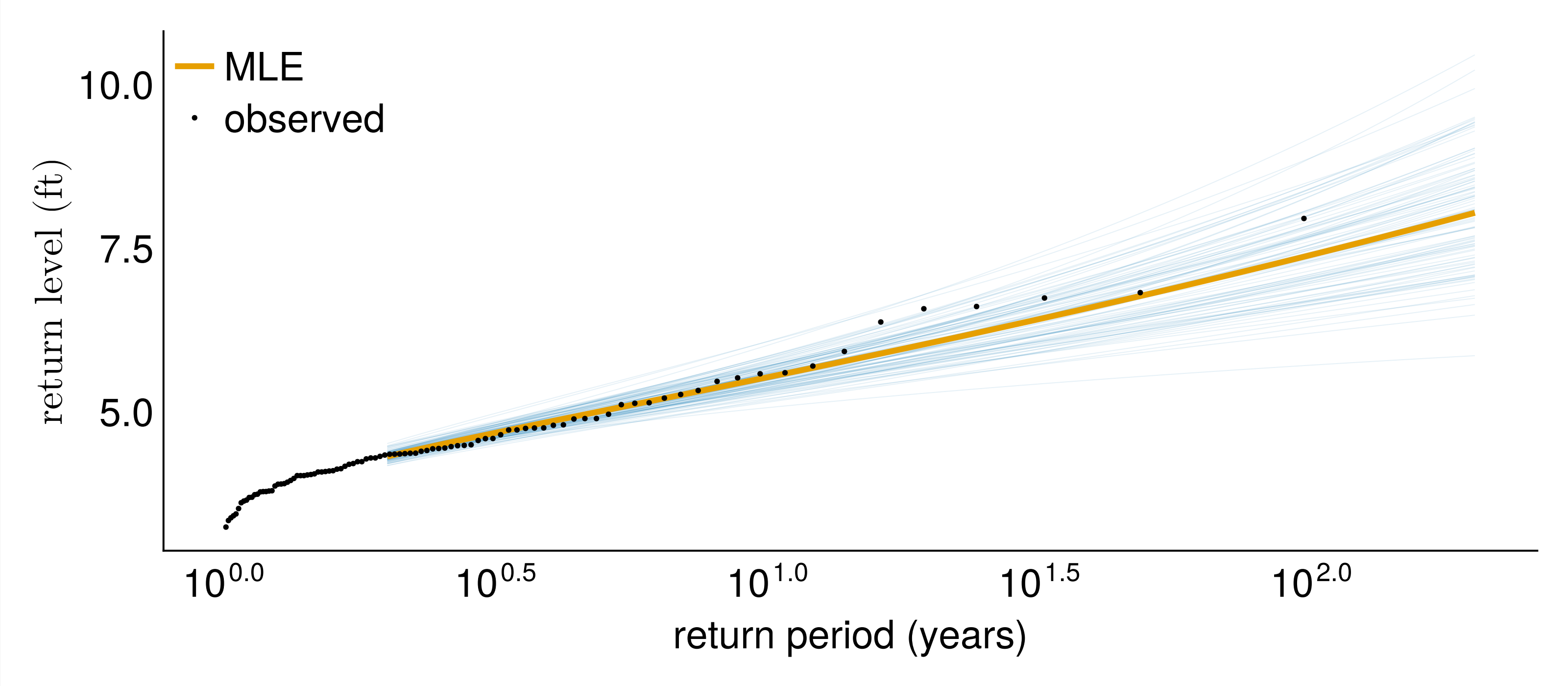
For supplementary notes, read Mudelsee (2010) §3.2 (pp. 74–76) on the bootstrap principle and Figure 3.3.

BOOTSTRAP THE RETURN LEVEL

Resample the Sewells Point annual maxima, fit a GEV on each resample, and compute the return-level curve $z_T = F^{-1}(1 - 1/T; \hat{\theta}^*)$:

```
n = n_obs
B = 100
return_periods = exp.(range(log(2), log(200); length=50))
boot_curves = zeros(B, length(return_periods))
for b in 1:B
    ys = y[rand(rng, 1:n, n)]
    d_b = only(getdistribution(gevfit(ys)))
    boot_curves[b, :] = [quantile(d_b, 1 - 1 / T) for T in return_periods]
end
```

BOOTSTRAP THE RETURN LEVEL



100 bootstrap resamples. The curves agree at short return periods and spread in the tail.

BOOTSTRAP VS BAYES

The bootstrap 95% CI on the 100-year return level is doing something that is conceptually *similar* to the Bayesian posterior:

```
MLE:        7.40 ft
Bootstrap: [6.36, 8.76] ft  (100 resamples)
Bayes:     [6.58, 9.44] ft  (16,000 posterior draws)
```

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THE ORDINARY BOOTSTRAP ASSUMES EXCHANGEABILITY

The Sewells Point bootstrap resampled individual years with replacement. That assumes any reordering of the data is equally likely. When the data has structure the resampling does not preserve, the confidence intervals are wrong.

AUTOCORRELATED DATA: THE BLOCK BOOTSTRAP

Annual temperature anomalies are positively autocorrelated: a warm year is more likely to follow a warm year. Resampling individual years destroys that dependence and makes the confidence interval too narrow (Mudelsee, 2010, §3.3).

The block bootstrap resamples **blocks** of l consecutive values, preserving the local dependence.

Parametric model: the parametric bootstrap

If you fitted a distribution by MLE, you can resample from the fitted distribution $\hat{F}(\hat{\theta})$, re-estimate on each synthetic sample, and look at the spread.

This is Monte Carlo from week 1, except the distribution itself is uncertain. With a Bayesian model you get this from the posterior predictive; the parametric bootstrap is the frequentist version.

LAB 3

Houston Hobby Airport, 88 years of annual minimum temperatures. The 2021 freeze reached -9.3 °C: how rare is that, and how sure are you?

1. Fit a Normal by MLE and by Bayes (conjugate prior)
2. Bootstrap the return period of a -9.3 °C winter
3. Compare the three uncertainty estimates side by side